

TP INFORMATIQUE N° 9

CORRECTION

Premiers tests

2) Une aile d'avion

Attention il faut annuler la dérivée

```
def f(x):  
    return(0.15*(3.7*x**0.5-3.4*x-0.3*x**4))  
  
def fp(x):  
    return(0.15*(0.5*3.7*x**(-0.5)-3.4-1.2*x**3))  
  
a=sp.newton(fp,0.5)  
  
print('La position du minimum est',a, 'ce qui donne une demi épaisseur de ',f(a))
```

3) une suite implicite

```
n = 100  
x = [k for k in range(n)]  
  
t = []  
for k in range(n):  
    xk = sp.bisect(lambda x : x**k*np.log(x)-1,1,3)  
    t.append(xk)  
  
pl.plot(x,t)  
pl.show()
```

Applications

4 Un polynôme

1. $c=0$
def f(x):
 return(x**3+c*x+1.)
def fp(x):
 return(3.*x**2+c)
2. pour $c > 0$ $f'(x) \Rightarrow 0$, la convergence est rapide.
3. pour $c = 0$ on a $f'(0) = 0$, ce qui pose problème autour de 0.
4. C'est pire, f' s'annule en deux points!

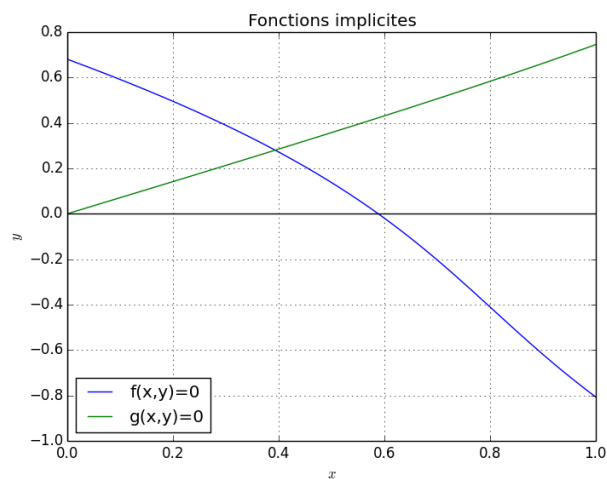
5) Fonction implicites

```
def f(x,y):
    return(np.exp(x+y**2)*np.sin(x+y)-1)
def g(x,y):
    return(np.cos(x)+np.sin(y)**2-1)

absc = np.linspace(0,1,200)
ordf=[]
for x in absc:
    ordf.append(sp.newton(lambda y :f(x,y),0.5))

ordg=[]
for x in absc:
    ordg.append(sp.newton(lambda y :g(x,y),0.5))
f,=pl.plot(absc,ordf)
g,=pl.plot(absc,ordg)
```

Et la figure :



6) Pseudo Newton

Pour la 4 :

```
def nl(A):
    return(max([sum([abs(A[i,j]) for i in range(len(A)) for j in range(len(A[0]))]))])

def nc(A):
    return(max([sum([abs(A[i,j]) for j in range(len(A[0])) for i in range(len(A))]))])

def pseunew(A,eps):
    X = 1/(nc(A)*nl(A))*np.transpose(A)
    while nc(np.dot(A,X)-np.eye(len(A)))>eps:
        X = np.dot(X,2*np.eye(len(A))-np.dot(A,X))
        #print(X)
    return(X)

A = np.array([[1,2,3],[4,5,6],[7,8,10]])

B =pseunew(A,10**-6)

print(np.dot(A,B))
```